F79SU Survival Models

Tutorial 6B: Graduation and Statistical Tests

1. The following table shows the actual deaths d_x, graduated mortality rates $\dot{q}_x$ and expected deaths $E_x\dot{q}_x$ on the basis of these graduated rates, from an investigation of life office assured lives mortality at policy durations 3 years and over. Some further information is also provided.

Ages Attained	d_x	$\dot{q}_x$	$E_x\dot{q}_x$	$d_x - E_x\dot{q}_x$	$\dfrac{d_x - E_x\dot{q}_x}{\sqrt{E_x\dot{q}_x(1-\dot{q}_x)}}$	$E_x\dot{q}_x(1-\dot{q}_x)$
15.5–19.5	63	0.00224	65	− 2	−0.248	64.9
20.5–24.5	668	0.00239	656	+ 12	+0.469	654.4
25.5–29.5	1505	0.00232	1533	− 28	−0.716	1529.4
30.5–34.5	2325	0.00261	2317	+ 8	+0.166	2311.0
35.5–39.5	3725	0.00331	3759	− 34	−0.555	3746.6
40.5–44.5	5733	0.00466	5591	+142	+1.904	5564.9
45.5–49.5	7618	0.00622	7734	−116	−1.323	7685.9
50.5–54.5	9953	0.00952	9930	+ 23	+0.232	9835.5
55.5–59.5	11801	0.01522	11806	− 5	−0.046	11626.3
60.5–64.5	12491	0.02516	12391	+100	+0.909	12079.2
65.5–69.5	13526	0.04045	13746	−220	−1.916	13190.0
70.5–74.5	15607	0.06732	15570	+ 37	+0.307	14521.8
75.5–79.5	14159	0.10341	14161	− 2	−0.018	12696.6
80.5–84.5	9183	0.15317	9149	+ 34	+0.386	7747.6
85.5–89.5	3919	0.21223	3931	− 12	−0.216	3096.7
90.5–94.5	999	0.28085	1008	− 9	−0.334	724.9
95.5–99.5	122	0.30847	145	− 23	−2.297	100.3
≥ 100.5	9	0.28800	15	− 6	−1.836	10.7

You may assume that the graduation process required 4 parameters to be determined.

By carrying out

(a) the χ^2 test,

(b) the standardized deviations test,

(c) the sign test,

(d) the change of sign test,

(e) the cumulative deviations test (*optional since not covered in this course*),

(f) the grouping of signs test, and

(g) the lag–1 serial correlation test

test the adherence to the data of this graduation. State, with reasons, whether you consider the graduated rates suitable for the calculation of premium rates and reserves for endowment assurances and whole life assurances.

2. The following data relate to a graduation which has been carried out by the graphical method.

Ages	E_x	$1000 \times \dot{q}_x$	$E_x\dot{q}_x$	d_x	$d_x - E_x\dot{q}_x$	$\frac{d_x - E_x\dot{q}_x}{\sqrt{E_x\dot{q}_x\dot{p}_x}}$
10-14	8119	1.747	14.18	14	−0.18	−0.048
15-19	7750	2.112	16.37	20	3.63	+0.898
20-24	6525	3.028	19.75	22	2.25	+0.506
25-29	5998	3.698	22.18	23	0.82	+0.174
30-34	5586	4.796	26.79	26	−0.79	−0.153
35-39	5245	5.248	27.52	28	0.48	+0.091
40-44	4659	6.806	31.71	32	0.29	+0.052
45-49	4222	8.915	37.64	37	−0.64	−0.105
50-54	3660	11.679	42.74	44	1.26	+0.193
55-59	3012	19.644	59.17	54	−5.17	−0.678
60-64	2500	28.725	71.81	68	−3.81	−0.456
65-69	2113	39.955	84.43	87	2.57	+0.286
70-74	1469	69.252	101.73	100	−1.73	−0.178
75-79	883	110.803	97.74	95	−2.84	−0.304
80-84	418	180.866	75.60	70	−5.60	−0.711
85+	181	292.293	52.90	49	−3.90	−0.638

(a) Explain the difficulties in carrying out a χ^2-test of a graphical graduation (*optional since not covered in this course*).

(b) The χ^2-test may fail to detect certain defects in a graduation. For each of the following suggest one test which would detect the defect, and carry it out.

(i) Fails to detect an abnormally large number of small deviations offset by a few large deviations.

(ii) Fails to detect positive or negative bias in the residuals.

(iii) Fails to detect clumping or non-random grouping of positive or negative signs.

(c) Test the smoothness of the graduation by considering third differences $\Delta^3\dot{q}_x$. What conclusion do you come to? (*optional since not covered in this course*)

On the basis of all the tests you have performed state with reasons whether or not you consider the graduation to be acceptable.